

Mennatullah Samir Mahmoud — Profile Knowledge Base

This document is a complete, structured reference about Mennatullah Samir Mahmoud, intended as context/knowledge-base source for a chatbot or RAG system answering questions on her behalf.

Basic Info

- **Full name:** Mennatullah Samir Mahmoud (also written "Menna Tullah Samir" / "Menna Allah Samir Mahmoud")
- **Location:** Giza, 6th of October, Egypt
- **Phone:** +20 1212391578
- **Email:** mntallhmstfy668@gmail.com
- **LinkedIn:** <https://www.linkedin.com/in/menna-tullah-samir-121a1b2b7/>
- **GitHub:** <https://github.com/mntallhmstfy668-sys>
- **Languages:** Arabic (Native), English (Good), French (Basic)
- **Current headline:** Software Engineer | AI Enthusiast | Computer Science Student @ Benha University | STEM Graduate

Summary / Bio

Computer Science and Artificial Intelligence undergraduate at Benha University with a strong passion for Artificial Intelligence, AI automation, and intelligent systems. Graduated from Sharkya STEM School, where she built a foundation in analytical thinking and solving complex problems. Currently focused on building automated workflows, multi-API integrations, and deploying Retrieval-Augmented Generation (RAG) pipelines and LLM-based solutions using Python, n8n, and LangChain. Interested in bridging automation and intelligent systems to create efficient, real-world solutions. Open to freelance work, collaborations, and hands-on opportunities in AI and automation.

Education

- **Bachelor of Science in Computers and Artificial Intelligence** – Faculty of Computers and Artificial Intelligence, Benha University, Cairo, Egypt. July 2025 – July 2029 (currently enrolled)
- **STEM Graduate** – Sharkya STEM School, Egypt. July 2022 – July 2025

Technical Skills

AI & Machine Learning: Natural Language Processing (NLP), Supervised Learning, Deep Learning, Scikit-learn, PyTorch, LLM Prompt Engineering, Text Classification, Data Cleaning & Vectorization

Generative AI / LLM Tooling: LLM agents & tool-calling, Retrieval-Augmented Generation (RAG), LangChain, Ollama (local LLM inference), ChromaDB (vector store), sentence-transformers (embeddings)

Automation & Integrations: n8n, API integrations, multi-API workflow automation

App / Interface Frameworks: Streamlit, Gradio

Mobile Development: Flutter, Dart

Programming Languages: Python, C++, Dart

Dev Tools: Git, GitHub, VS Code, Google Colab

Embedded & Hardware: Arduino

Certifications

Certificate	Issuer	Date
AI & Machine Learning Core Concepts and Practical Skills	Neurova AI Bootcamp	May 2026
Introduction to C++ (Course Certificate)	SoloLearn	Dec 2025
Robotics with Arduino (Certificate of Accomplishment)	Robotix	Aug 3, 2023
Certificate of Completion - Ramadan Program (Flutter Track)	TechT	-
Digital Literacy Certification	-	-

Training in progress (not yet certified)

AI Automation Track, Digital Egypt Pioneers Initiative (DEPI) – Ministry of Communications and Information Technology (MCIT). Jul 2026 – Present. Intensive training in AI workflow automation (n8n), API integrations, and prompt engineering strategies. *(Certificate not issued yet – do not present as completed.)*

Projects

1. AI Research Assistant (RAG & Gradio)

Lead AI Developer Repo: https://github.com/mntallhmstfy668-sys/Ai_Research_Assistant-

An autonomous research assistant. Given a research topic, the language model (Ollama, qwen2.5:7b) decides — via tool-calling — which of six tools to invoke and in what order: web search, page scraping, source summarization, source comparison, report generation, and querying a RAG memory store. Produces a structured report (Introduction, Key Findings, Conclusion, Sources) and supports grounded follow-up Q&A over the sources already collected.

- **Tech stack:** Python, Ollama (qwen2.5:7b), Gradio, ChromaDB, sentence-transformers (all-MiniLM-L6-v2), LangChain's RecursiveCharacterTextSplitter, ddgs (DuckDuckGo search), requests + BeautifulSoup4
- **How it works:** Chunking → embedding → storage in ChromaDB → semantic retrieval for follow-up questions → grounded generation (retrieved chunks injected into the prompt).
- **Reliability:** input validation, per-tool error handling with graceful fallback, explicit message if the agent can't complete within its step limit.
- **Result / impact:** Orchestrates 6 tools autonomously (search → scrape → summarize → compare → report) with zero hard-coded workflow logic — the model itself decides the sequence.

2. CV & Job Description Matcher (AI Resume & Portfolio Job Matcher)

Lead AI Developer Repo: <https://github.com/mntallhmstfy668-sys/Resume-Analyzer>

A two-pass local LLM pipeline (Ollama, Llama 3.1:8b) that ingests a candidate's resume (raw text, PDF, DOCX) or portfolio URL, extracts a structured JSON candidate profile (Pass 1), then compares that profile against a target job description to generate a match report with match score, strengths, and skill gaps (Pass 2).

- **Tech stack:** Python, Streamlit, Ollama (Llama 3.1:8b) via OpenAI-compatible SDK, BeautifulSoup4, requests, python-docx
- **Runs 100% locally** — full data privacy, no external API costs.
- **Result / impact:** One run takes a raw resume (PDF, DOCX, or URL) all the way to a scored job-match report.

3. AI-Powered STEM Student Chatbot Assistant

Lead AI Developer Repo: <https://github.com/mntallhmstfy668-sys/GPT-4o-ChatBot-using-Streamlit>

A conversational assistant that explains STEM concepts in plain language, using structured LLM prompt engineering and dialogue-flow logic with reusable, curriculum-aligned prompt templates.

- **Tech stack:** Python, Streamlit, GPT-4o, prompt engineering
- **Result / impact:** Adapts each explanation live to the student's exact question instead of one fixed script for everyone.

4. Fruits Classification System

Computer Vision Developer Repo: <https://github.com/mntallhmstfy668-sys/fruit-classifier>

A supervised, multi-class image classification pipeline that categorizes a fruit image dataset: data preprocessing, feature extraction, model training, and evaluation with standard classification metrics.

- **Tech stack:** Python, Scikit-learn, image processing
- **Result / impact:** Full pipeline — raw image dataset → feature extraction → trained classifier → evaluated with standard metrics.

5. Weather Station & Eco-Friendly Monitoring System

STEM Project Developer

An environmental monitoring system combining hardware sensors with C++ firmware logic; real-time routines collect and process temperature, humidity, and gas concentration data.

- **Tech stack:** C++, Arduino, hardware sensors

6. Hydroponics & Smart Irrigation Control System

STEM Project Developer

An automated smart-agriculture prototype using sensor feedback loops to monitor water and nutrient levels; embedded programming logic reduces resource waste and improves system responsiveness.

- **Tech stack:** C++, Arduino, sensors, embedded programming

Availability & What She's Looking For

Open to **freelance projects and collaborations** in AI, automation, and software development — including AI agents/RAG systems, chatbots, workflow automation (n8n), and practical machine learning tools. Comfortable delivering either cloud-based or fully local/private solutions (no data leaving the client's machine, no ongoing API costs) depending on the client's needs. Also open to internships/entry-level roles and research collaborations.

Contact

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